

Report

Contents

1	Introduction	2
2	Probability Spaces	3
3	Random Variables	4
	Types of Random Variables	4
	Distribution of a Random Variable	5
	Transformations	5
	Example (Inverse Transform Sampling)	5
4	Stochastic Processes	5
	Types of Stochastic Processes	6
	Sample Paths and Adaptation	6
	Filtration	6
	Stationarity	6
	Examples in Finance	6
5	Martingales	7
	Submartingales and Supermartingales	7
	Examples	7
6	Brownian Motion	8
	Definition	8
	Properties	8
7	Itô Integral and Ito Process	9
	Motivation	9
	Definition	9
	Key Properties	10
	Example	10
	Definition of Itô Process	10

8	Itô's Formula	11
	One-Dimensional Case	11
	Interpretation	11
	Example	12
	Applications in Finance	12
9	Stochastic Differential Equations (SDEs)	12
	Existence and Uniqueness	12
	Example: Geometric Brownian Motion (GBM)	12
	Applications in Finance	13
10	Market Models	14
	The Risky Asset	15
	The Risk-Free Interest Rate	15
	The Risk-Free Asset	15
	Discounting Process	15
	Summary of the Market Model	15
11	The Black–Scholes Model	16
	Market Assumptions	16
	No-Arbitrage and the Risk-Neutral Measure	17
	Pricing a European Option	17
	Closed-Form Solution (Black–Scholes Formula)	17
	Remarks and Limitations	18
12	Conclusion	20
	Proof of Itô Isometry	21
	Derivation of Black–Scholes PDE	21
	Derivation of Geometric Brownian Motion Solution	22
	Normal Distribution Identities	22
	Greeks Formulas Summary	22

1 Introduction

Financial markets are inherently uncertain, influenced by a wide range of unpredictable factors such as news events, investor behavior, and macroeconomic indicators. To model this uncertainty in a rigorous and realistic way, stochastic calculus has become a central mathematical tool in modern finance. It allows us to describe the random evolution of asset prices, assess risk, and derive pricing formulas for complex derivatives.

The purpose of this report is to study how stochastic calculus can be applied to financial modeling. To reach this objective, we begin by introducing the fundamental mathematical framework, including probability spaces, random variables, and stochastic processes. Building upon these foundations, we will explore core concepts such as martingales, Brownian motion, Itô integrals, Itô processes, and Itô's formula. The final sections will apply these tools to

financial market models, culminating in key results such as the Black–Scholes formula and the analysis of option sensitivities, known as the Greeks.

2 Probability Spaces

Definition 2.1 *Hello*

2.1

A probability space is the mathematical foundation for modeling uncertainty. It provides a formal structure for describing all possible outcomes of a random experiment and assigning probabilities to events. A probability space is defined as a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where:

- **Sample space** Ω : the set of all possible outcomes of a random experiment. For example, in modeling asset prices, Ω could represent all possible paths a stock price can take over time.
- **Sigma-algebra** $\mathcal{F} \subseteq 2^\Omega$: a collection of subsets of Ω (called events) that satisfies:
 - $\Omega \in \mathcal{F}$ (contains the whole space),
 - If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ (closed under complementation),
 - If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$ (closed under countable unions).

The sigma-algebra defines the events to which we can assign probabilities.

- **Probability measure** $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$: a function that assigns probabilities to events in $\mathcal{F}$, satisfying:
 - $\mathbb{P}(\Omega) = 1$,
 - $\mathbb{P}(\emptyset) = 0$,
 - Countable additivity: if $A_1, A_2, \dots \in \mathcal{F}$ are disjoint, then

$$\mathbb{P} \left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

This structure allows us to model randomness in a rigorous way. In financial applications, the sample space may represent all possible future evolutions of asset prices, the sigma-algebra captures observable events (e.g., a stock price hitting a barrier), and the probability measure quantifies the likelihood of such events, often under a risk-neutral or real-world probability.

As we move toward stochastic processes, we will use this formalism to define random variables, filtrations, and martingales in a consistent probability framework.

3 Random Variables

A random variable is a measurable function that assigns a real number to each outcome in a probability space. It is the main tool for quantifying uncertainty in probabilistic models.

Formally, given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a function

$$X : \Omega \rightarrow \mathbb{R}$$

is called a **random variable** if it is **measurable**, meaning that for every Borel set $B \subseteq \mathbb{R}$, the preimage $\{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$. This ensures that the event “ $X \in B$ ” is well-defined and has a probability.

In practice, it suffices to verify that the set $\{X \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. This leads to the definition of the **cumulative distribution function (CDF)**:

$$F_X(x) = \mathbb{P}(X \leq x), \quad x \in \mathbb{R}.$$

Types of Random Variables

Random variables can be classified as:

- **Discrete**, if they take values in a countable set. Examples include:
 - Bernoulli random variable (e.g., success/failure),
 - Binomial, Poisson, or Geometric distributions.
- **Continuous**, if they are supported on an interval or subset of $\mathbb{R}$, and have a **probability density function (PDF)** f_X such that:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

Examples include the Uniform, Exponential, and Normal distributions.

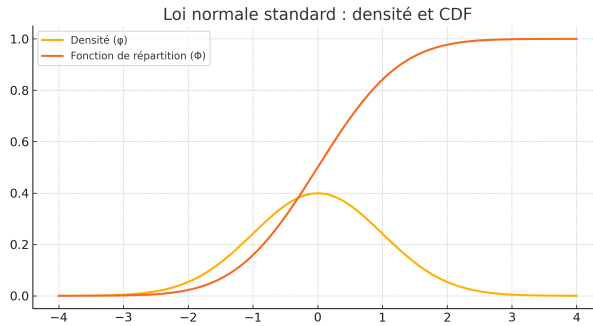


Figure 1: Density and cumulative distribution function of the normal distribution

Distribution of a Random Variable

Given a random variable X , we define its **law** or **distribution** as a probability measure μ_X on $\mathbb{R}$ by:

$$\mu_X(B) = \mathbb{P}(X \in B) \quad \text{for all Borel sets } B.$$

This allows us to transfer probability measures from Ω to the real line, where analysis is easier.

Transformations

If $Y = g(X)$ for some measurable function g , then Y is also a random variable. Its distribution can often be found using:

- The CDF method: compute $F_Y(y) = \mathbb{P}(g(X) \leq y)$,
- Change of variables if g is differentiable and monotonic.

Example (Inverse Transform Sampling)

A fundamental technique in simulation is the inverse transform method. If $U \sim \text{Uniform}[0, 1]$ and F_X is a continuous and strictly increasing CDF, then:

$$X = F_X^{-1}(U)$$

has distribution F_X . This method is widely used in Monte Carlo simulations and financial modeling.

Random variables serve as the building blocks for defining more complex objects like stochastic processes, where the value of the variable evolves over time in a random way, a key idea in modeling financial markets.

4 Stochastic Processes

A stochastic process is a collection of random variables indexed by time, used to model the evolution of systems under uncertainty. In finance, stochastic processes are essential tools for modeling quantities such as asset prices, interest rates, and volatility over time.

Formally, let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A **stochastic process** is a family of random variables $\{X_t\}_{t \in T}$, where each

$$X_t : \Omega \rightarrow \mathbb{R}$$

is a random variable, and $T \subseteq \mathbb{R}_+$ is an index set, typically representing time.

Types of Stochastic Processes

- **Discrete-time:** The index set is $T = \mathbb{N}$ or $\{0, 1, 2, \dots\}$, e.g., daily stock prices.
- **Continuous-time:** The index set is $T = [0, \infty)$, which is used for modeling continuous market dynamics, such as in the Black–Scholes framework.

Sample Paths and Adaptation

For each $\omega \in \Omega$, the function $t \mapsto X_t(\omega)$ is called the **sample path** of the process. It represents a possible realization of the evolution of the random quantity over time.

A process $\{X_t\}_{t \geq 0}$ is said to be **adapted** to a filtration $\{\mathcal{F}_t\}_{t \geq 0}$ if for each t , the random variable X_t is $\mathcal{F}_t$ -measurable. This means that the value of X_t depends only on information available up to time t , ensuring no "anticipation" of future randomness.

Filtration

A **filtration** $\{\mathcal{F}_t\}_{t \geq 0}$ is an increasing family of sub- σ -algebras of $\mathcal{F}$:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}, \quad \text{for all } 0 \leq s \leq t.$$

It models the accumulation of information over time. In financial contexts, $\mathcal{F}_t$ represents all the information available to market participants up to time t .

Stationarity

A stochastic process $\{X_t\}$ is said to be **strictly stationary** if, for any finite collection of time points $t_1, \dots, t_n$ and for any $h \geq 0$, the joint distribution of $(X_{t_1}, \dots, X_{t_n})$ is the same as that of $(X_{t_1+h}, \dots, X_{t_n+h})$.

In finance, strict stationarity is rarely assumed for asset prices, which tend to trend over time. However, certain quantities such as log returns or volatility may be assumed to be (weakly) stationary for modeling purposes.

Examples in Finance

- S_t : The price of a stock at time t ,
- R_t : The accumulated return up to time t ,
- V_t : The volatility of an asset over time,
- X_t : The short-term interest rate in term structure models.

Stochastic processes are the backbone of modern quantitative finance. In the next sections, we will study specific types of processes, such as martingales and Brownian motion, that are essential for pricing derivatives and modeling arbitrage-free markets.

5 Martingales

Martingales are a special class of stochastic processes that model "fair games" where, conditional on the past, the expected future value is equal to the current value. They play a central role in stochastic calculus and financial mathematics, particularly in arbitrage theory and risk-neutral pricing.

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ be a filtered probability space. A stochastic process $\{M_t\}_{t \geq 0}$ is called a **martingale** with respect to the filtration $\{\mathcal{F}_t\}$ if it satisfies the following conditions:

- M_t is $\mathcal{F}_t$ -measurable for all $t \geq 0$ (i.e., the process is adapted),
- $\mathbb{E}[|M_t|] < \infty$ for all $t \geq 0$ (integrability),
- $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$ for all $0 \leq s \leq t$ (martingale property).

This means that the best prediction of the future value M_t , given all the information up to time s , is simply the current value M_s . In this sense, martingales have no "drift" and represent models where there is no predictable trend.

Submartingales and Supermartingales

The martingale condition can be weakened to define:

- A process $\{X_t\}$ is a **submartingale** if

$$\mathbb{E}[X_t | \mathcal{F}_s] \geq X_s, \quad \forall s \leq t.$$

- A process $\{X_t\}$ is a **supermartingale** if

$$\mathbb{E}[X_t | \mathcal{F}_s] \leq X_s, \quad \forall s \leq t.$$

These notions are useful for modeling processes with upward or downward trends in expectation, such as growing risk or decaying asset values.

Examples

- A symmetric random walk is a discrete-time martingale.
- The discounted price of a financial asset under a risk-neutral measure (i.e., $\tilde{S}_t = e^{-rt} S_t$) is a martingale in arbitrage-free markets.
- If B_t is a standard Brownian motion, then B_t is a continuous-time martingale.
- For Brownian motion B_t , the process $B_t^2 - t$ is also a martingale.

Martingale path vs non martingale

In the next section, we introduce Brownian motion, the most important continuous-time stochastic process, and the building block of stochastic calculus.

6 Brownian Motion

Brownian motion is the fundamental continuous-time stochastic process in probability theory and the cornerstone of modern mathematical finance. It models the random evolution of asset prices, interest rates, and other financial variables, and serves as the driving force behind stochastic integrals and differential equations.

Definition

A stochastic process $\{B_t\}_{t \geq 0}$ is called a **standard Brownian motion** (or **Wiener process**) if it satisfies the following properties:

- $B_0 = 0$ almost surely,
- $\{B_t\}$ has independent increments: for any $0 \leq s < t$, the increment $B_t - B_s$ is independent of $\mathcal{F}_s$,
- $\{B_t\}$ has stationary increments: the distribution of $B_t - B_s$ depends only on $t - s$,
- $B_t - B_s \sim \mathcal{N}(0, t - s)$: normally distributed with mean 0 and variance $t - s$,
- The sample paths $t \mapsto B_t(\omega)$ are continuous almost surely.

Properties

- $\mathbb{E}[B_t] = 0$, $\text{Var}(B_t) = t$,
- Brownian motion is a martingale: $\mathbb{E}[B_t \mid \mathcal{F}_s] = B_s$,
- Brownian motion has infinite variation but finite quadratic variation:

$$[B]_t = \lim_{\max \Delta t_i \rightarrow 0} \sum_i (B_{t_{i+1}} - B_{t_i})^2 = t.$$

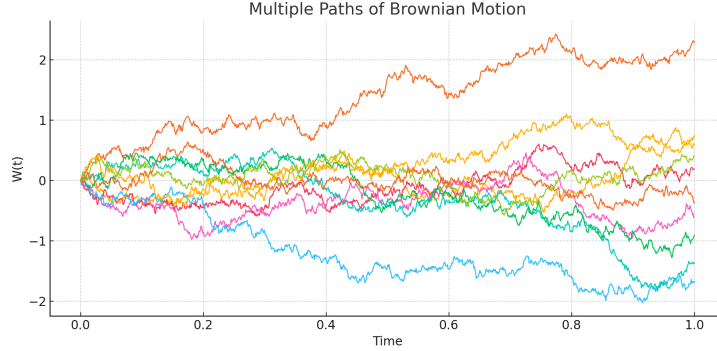


Figure 2: Simulation of a Brownian motion

In the following sections, we will construct integrals with respect to Brownian motion (Itô integrals) and study how to model stochastic dynamics using Itô calculus.

7 Itô Integral and Ito Process

The Itô integral extends the notion of integration to allow integration with respect to Brownian motion and other martingales, processes that are nowhere differentiable and have infinite variation. This is essential for defining and solving stochastic differential equations used in financial modeling.

Motivation

In classical calculus, integrals of the form $\int_0^t f(s) ds$ represent accumulated deterministic changes. In stochastic settings, however, we often encounter expressions like:

$$\int_0^t H_s dB_s,$$

where B_s is Brownian motion and H_s is a stochastic (random) process. Since Brownian motion has unbounded variation, classical integration methods like Riemann or Riemann–Stieltjes fail. The Itô integral provides a consistent and rigorous way to define such expressions.

Definition

Let $\{H_t\}_{t \geq 0}$ be an adapted process that is square-integrable, meaning:

$$\mathbb{E} \left[\int_0^T H_t^2 dt \right] < \infty.$$

Then the **Itô integral** of H with respect to Brownian motion $\{B_t\}$ is defined as a limit in probability of Riemann sums:

$$\int_0^T H_t dB_t = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} H_{t_i} (B_{t_{i+1}} - B_{t_i}),$$

where the partition $\{t_i\}$ of $[0, T]$ becomes finer as $n \rightarrow \infty$, and the integrand is evaluated at the left endpoint (making it non-anticipative).

Key Properties

- **Linearity:** $\int_0^t (aH_s + bG_s) dB_s = a \int_0^t H_s dB_s + b \int_0^t G_s dB_s$.
- **Martingale property:** If H_t is adapted and square-integrable, then $\int_0^t H_s dB_s$ is a martingale.
- **Itô Isometry:**

$$\mathbb{E} \left[\left(\int_0^T H_s dB_s \right)^2 \right] = \mathbb{E} \left[\int_0^T H_s^2 ds \right].$$

This powerful identity allows us to compute variances of stochastic integrals directly from the integrand.

Example

Let $H_t = B_t$. Then:

$$\int_0^t B_s dB_s = \frac{1}{2} B_t^2 - \frac{1}{2} t.$$

This result comes from applying Itô's formula and reflects the fact that stochastic integration includes a correction term due to the randomness of the integrator.

An Itô process is a type of stochastic process defined as the solution to a stochastic differential equation (SDE). These processes extend classical differential equations by incorporating randomness through Brownian motion. In financial mathematics, they are used to model the evolution of asset prices, interest rates, and volatility under uncertainty.

Definition of Itô Process

A stochastic process $\{X_t\}_{t \geq 0}$ is called an **Itô process** if it can be written as:

$$X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dB_s,$$

where:

- $X_0 \in \mathbb{R}$ is the initial value,
- μ_t is the drift term (predictable, integrable),
- σ_t is the diffusion (volatility) term (adapted, square-integrable),
- B_t is standard Brownian motion.

In differential notation, this is often written as:

$$dX_t = \mu_t dt + \sigma_t dB_t.$$

8 Itô's Formula

Itô's formula is the stochastic counterpart of the chain rule in classical calculus. It provides a method for computing the differential of a function applied to a stochastic process, particularly one driven by Brownian motion. It is one of the most powerful tools in stochastic calculus and plays a central role in financial modeling, including the derivation of the Black-Scholes equation.

One-Dimensional Case

Let X_t be an Itô process of the form:

$$dX_t = \mu_t dt + \sigma_t dB_t,$$

and let $f(t, x)$ be a function that is once differentiable in time and twice differentiable in space ($C^{1,2}$). Then, Itô's formula states:

$$df(t, X_t) = \frac{\partial f}{\partial t}(t, X_t) dt + \frac{\partial f}{\partial x}(t, X_t) dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t) \sigma_t^2 dt.$$

Substituting dX_t , we get:

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{1}{2} \sigma_t^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma_t \frac{\partial f}{\partial x} dB_t.$$

Interpretation

The key feature of Itô's formula is the appearance of the second derivative term, which accounts for the randomness and quadratic variation of Brownian motion. This correction term distinguishes Itô calculus from classical differential calculus.

Example

Let $X_t = B_t$ and $f(x) = x^2$. Then applying Itô's formula:

$$df(B_t) = 2B_t dB_t + \frac{1}{2}(2) dt = 2B_t dB_t + dt.$$

Integrating both sides from 0 to t , we get:

$$B_t^2 = 2 \int_0^t B_s dB_s + t.$$

Applications in Finance

Itô's formula is used to compute the dynamics of functions of stochastic processes, such as the logarithm of a stock price or the value of a derivative security. In the Black-Scholes model, it is essential for deriving the partial differential equation that the option price must satisfy.

9 Stochastic Differential Equations (SDEs)

A stochastic differential equation (SDE) models the evolution of a process whose infinitesimal changes involve both a deterministic trend and a random component driven by Brownian motion. The general form is:

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dB_t,$$

with initial condition $X_0 = x \in \mathbb{R}$.

Existence and Uniqueness

Under suitable conditions (e.g., Lipschitz continuity and linear growth of μ and σ), the SDE admits a unique strong solution. The theory ensures that we can define a stochastic process X_t that evolves consistently with the given dynamics.

Example: Geometric Brownian Motion (GBM)

A widely used model in finance is the geometric Brownian motion, which satisfies the SDE:

$$dS_t = \mu S_t dt + \sigma S_t dB_t.$$

Its solution is:

$$S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t \right).$$

This model captures the multiplicative and continuous nature of asset prices, and underpins the Black-Scholes model for option pricing.

Applications in Finance

Itô processes and SDEs are fundamental in modeling:

- Stock price dynamics (e.g., Black–Scholes model),
- Interest rate models (e.g., Vasicek, CIR),
- Volatility modeling (e.g., Heston model),
- Portfolio optimization and hedging strategies.

In the next section, we will apply these tools to derive the Black–Scholes formula and explore its implications for derivative pricing and hedging.

Derivatives and Payoff Structures

A financial derivative is a contract whose value depends on the performance of an underlying asset such as a stock, bond, commodity, or interest rate. Derivatives are used for hedging, speculation, and arbitrage, and they play a central role in modern financial markets.

Basic Types of Derivatives

- **European Call Option:** Gives the holder the right (but not the obligation) to buy the underlying asset at a fixed strike price K at maturity T . Its payoff is:

$$\max(S_T - K, 0).$$

- **European Put Option:** Gives the holder the right (but not the obligation) to sell the underlying asset at a fixed strike price K at maturity T . Its payoff is:

$$\max(K - S_T, 0).$$

- **Forward Contract:** An agreement to buy (or sell) an asset at a specified future time T for a predetermined price F . Unlike options, forwards are obligations, not rights. The payoff to the buyer at maturity is:

$$S_T - F,$$

while the seller receives:

$$F - S_T.$$

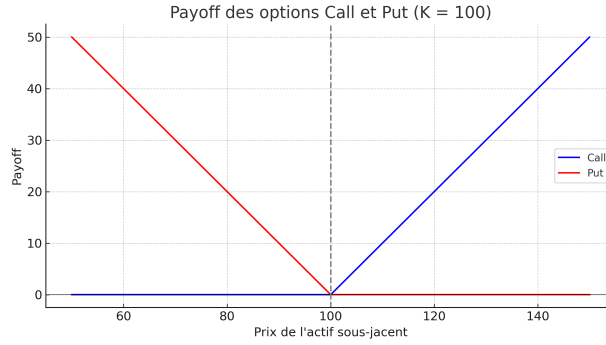


Figure 3: Payoff diagrams for European call and put options

European vs. American Options

In this report, we focus exclusively on European-style options, which can only be exercised at maturity T . In contrast, American options can be exercised at any time before or at maturity, making their valuation more complex and dependent on early exercise decisions.

Key Features of Derivatives

- Their value is derived from the underlying asset.
- They can be used to transfer risk between parties.
- Many derivatives have non-linear payoffs.
- Pricing requires probabilistic and no-arbitrage tools, especially in continuous-time models.

10 Market Models

A financial market model is a probabilistic framework describing the evolution of asset prices and interest rates over time. Formally, it is defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and consists of a pair of stochastic processes:

$$(S_t, t \geq 0) \quad \text{and} \quad (R_t, t \geq 0),$$

along with a filtration $(\mathcal{F}_t, t \geq 0)$ representing the flow of information over time.

The Risky Asset

The process (S_t) represents the price of a risky asset such as a stock, bond, commodity, or foreign currency. In continuous-time financial models, S_t is typically modeled as an Itô process, often satisfying an SDE of the form:

$$dS_t = \mu_t S_t dt + \sigma_t S_t dB_t,$$

where μ_t is the drift, σ_t the volatility, and B_t a Brownian motion.

The Risk-Free Interest Rate

The process (R_t) models the instantaneous interest rate available from a risk-free investment such as a money market account. In the simplest case, we assume a constant rate:

$$R_t = r, \quad \text{for all } t \geq 0.$$

The Risk-Free Asset

The value A_t of a money market account invested at the risk-free rate evolves according to:

$$dA_t = R_t A_t dt.$$

Solving this equation yields:

$$A_t = A_0 \exp \left(\int_0^t R_s ds \right).$$

If $R_t = r$ is constant, then $A_t = A_0 e^{rt}$. This asset is used as the benchmark for discounting future cash flows in a no-arbitrage pricing framework.

Discounting Process

To express the present value of future cash flows, we define the discounting process $(D_t, t \geq 0)$ as:

$$D_t = \exp \left(- \int_0^t R_s ds \right).$$

This quantity converts values at time t into equivalent values at time 0. For example, under constant interest rate $R_t = r$, we have $D_t = e^{-rt}$. The discounting process is essential in risk-neutral valuation and computing expected present values.

Summary of the Market Model

In summary, a market model consists of:

- A risky asset (S_t) , typically an Itô process,

- A short rate process (R_t) , possibly stochastic or constant,
- A risk-free asset (A_t) , growing according to the interest rate,
- A discount factor (D_t) , used to translate future values into present values.

This formal framework enables the consistent pricing of financial derivatives, which we will explore next using the Black–Scholes model.

11 The Black–Scholes Model

Building on the market model introduced in the previous section, we now turn to the Black–Scholes model, one of the most fundamental and celebrated frameworks in quantitative finance. It provides a closed-form solution for the price of European options under idealized but tractable assumptions.

Market Assumptions

The Black–Scholes model operates in a continuous-time market defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with a filtration $(\mathcal{F}_t)$ satisfying the usual conditions. The model includes two assets:

- A risk-free asset (A_t) , also known as the money market account, that grows at a constant interest rate $r \geq 0$:

$$dA_t = rA_t dt, \quad \Rightarrow \quad A_t = A_0 e^{rt}.$$

- A risky asset (S_t) , typically a stock, which follows a geometric Brownian motion:

$$dS_t = \mu S_t dt + \sigma S_t dB_t,$$

where $\mu \in \mathbb{R}$ is the expected return, $\sigma > 0$ is the volatility, and B_t is a standard Brownian motion under the physical measure $\mathbb{P}$.

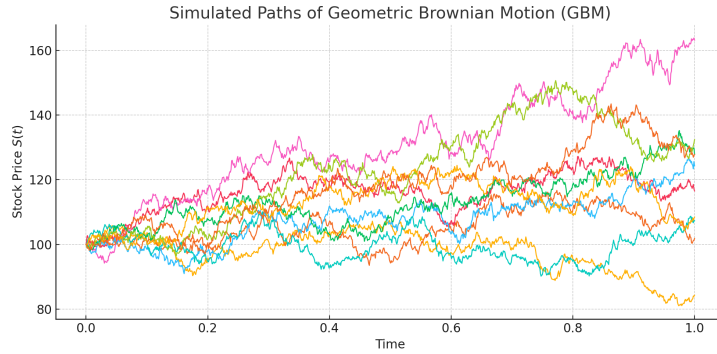


Figure 4: Simulated Paths of a Geometric Brownian Motion (GBM) for Stock Prices

No-Arbitrage and the Risk-Neutral Measure

To price derivatives consistently with no-arbitrage, we move to a probability measure $\mathbb{Q} \sim \mathbb{P}$, under which the discounted stock price $\tilde{S}_t = e^{-rt} S_t$ becomes a martingale. Under $\mathbb{Q}$, the stock evolves as:

$$dS_t = rS_t dt + \sigma S_t dW_t,$$

where W_t is a $\mathbb{Q}$ -Brownian motion (constructed via Girsanov's theorem). This shift reflects the idea that in a risk-neutral world, all assets grow on average at the risk-free rate.

Pricing a European Option

Let $C(t, S_t)$ be the price at time t of a European call option with maturity T and strike K . Under the risk-neutral measure, its price is given by the discounted expected payoff:

$$C(t, S_t) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}} [(S_T - K)^+ | \mathcal{F}_t].$$

Using Itô's formula and the Feynman-Kac theorem, one derives that $C(t, S_t)$ satisfies the Black-Scholes partial differential equation (PDE):

$$\frac{\partial C}{\partial t} + rS \frac{\partial C}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} = rC.$$

Closed-Form Solution (Black-Scholes Formula)

Solving the PDE with terminal condition $C(T, S_T) = (S_T - K)^+$, we obtain the celebrated Black-Scholes formula:

$$C(0, S_0) = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2),$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

and $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution.

Numerical Example: Black-Scholes Formula **Given:**

- Stock price $S = 100$
- Strike price $K = 100$
- Time to maturity $T - t = 1$ year
- Risk-free rate $r = 5\%$
- Volatility $\sigma = 20\%$

Step-by-step calculation:

$$d_1 = \frac{\ln(100/100) + (0.05 + \frac{1}{2} \cdot 0.2^2)}{0.2} = 0.35$$

$$d_2 = d_1 - 0.2 = 0.15$$

$$C = 100 \cdot \Phi(0.35) - 100 \cdot e^{-0.05} \cdot \Phi(0.15)$$

$$C \approx 100 \cdot 0.6368 - 95.12 \cdot 0.5596 = \mathbf{10.45}$$

Conclusion: The theoretical price of the call option is approximately **10.45**.

Remarks and Limitations

The Black–Scholes model is elegant and analytically tractable, but it relies on strong assumptions:

- Constant interest rate r and volatility σ ,
- Log-normal distribution of returns,
- No transaction costs or taxes,
- Continuous trading and perfect markets,
- No early exercise (for European options).

Despite these limitations, the Black–Scholes model remains a foundational tool in finance. It also serves as a benchmark for more advanced models that relax these assumptions.

In the next section, we explore the sensitivities of the Black–Scholes price with respect to its input parameters, known as the Greeks, which are crucial for risk management and hedging.

The Greeks

The **Greeks** are partial derivatives of the option price with respect to various input parameters in the Black–Scholes model. They quantify how sensitive the value of a derivative is to changes in underlying market variables, and are essential tools for risk management and hedging.

Delta (Δ)

Delta measures the sensitivity of the option price to changes in the underlying asset price:

$$\Delta = \frac{\partial C}{\partial S}.$$

For a European call option under the Black–Scholes model:

$$\Delta_{\text{call}} = \Phi(d_1), \quad \Delta_{\text{put}} = \Phi(d_1) - 1.$$

It represents the hedge ratio: the number of units of the underlying needed to replicate the option.

Gamma (Γ)

Gamma measures the rate of change of Delta with respect to the underlying asset price:

$$\Gamma = \frac{\partial^2 C}{\partial S^2}.$$

For both call and put options:

$$\Gamma = \frac{\phi(d_1)}{S\sigma\sqrt{T-t}},$$

where $\phi(\cdot)$ is the standard normal density function. Gamma helps assess the non-linearity of an option's value with respect to price changes.

Vega

Vega measures sensitivity to changes in volatility:

$$\text{Vega} = \frac{\partial C}{\partial \sigma} = S\phi(d_1)\sqrt{T-t}.$$

Vega is the same for calls and puts. It captures the impact of volatility changes on the option price and is especially important for options traders.

Theta (Θ)

Theta measures the sensitivity of the option price to the passage of time, also called time decay:

$$\Theta = \frac{\partial C}{\partial t}.$$

For a call option:

$$\Theta_{\text{call}} = -\frac{S\phi(d_1)\sigma}{2\sqrt{T-t}} - rKe^{-r(T-t)}\Phi(d_2).$$

Theta is usually negative, indicating that the value of an option decreases as time passes (all else equal).

Rho (ρ)

Rho measures sensitivity to the interest rate:

$$\rho = \frac{\partial C}{\partial r}.$$

For a call option:

$$\rho_{\text{call}} = K(T-t)e^{-r(T-t)}\Phi(d_2),$$

and for a put:

$$\rho_{\text{put}} = -K(T-t)e^{-r(T-t)}\Phi(-d_2).$$

Summary Table

Greek	Interpretation
Delta (Δ)	Sensitivity to asset price
Gamma (Γ)	Sensitivity of delta (convexity)
Vega	Sensitivity to volatility
Theta (Θ)	Sensitivity to time decay
Rho (ρ)	Sensitivity to interest rate

Applications

Greeks are used to construct dynamic hedging strategies. For example:

- **Delta hedging** involves holding a quantity Δ of the underlying asset to neutralize small price movements.
- **Gamma** and **Vega** are used to manage non-linear risk and volatility exposure.

With the Black–Scholes model and the Greeks in hand, practitioners have the basic toolkit to price, hedge, and analyze simple derivatives. However, real markets often deviate from model assumptions, motivating more advanced approaches, such as stochastic volatility models, jump processes, and numerical methods.

12 Conclusion

In this report, we explored the foundational tools of stochastic calculus and their applications in financial modeling. We began with a rigorous construction of probability spaces and stochastic processes, developed key concepts such as Brownian motion, Itô integrals, and stochastic differential equations (SDEs), and applied these tools within the framework of market models.

The Black–Scholes model provided a concrete and elegant example of how stochastic calculus can be used to derive pricing formulas for financial derivatives. Through the derivation of the Black–Scholes PDE and its closed-form solution, we demonstrated how concepts like risk-neutral valuation, no-arbitrage

pricing, and martingale measures come together to form a powerful pricing framework.

We also introduced the Greeks, which quantify the sensitivities of option prices to market parameters and form the basis for hedging and risk management strategies.

While the Black–Scholes model is a cornerstone of mathematical finance, it relies on simplifying assumptions such as constant volatility and frictionless markets. Real-world financial markets often exhibit features like stochastic volatility, jumps, illiquidity, and market imperfections. These limitations motivate the development of more advanced models, such as the Heston model, Lévy processes, and numerical techniques like Monte Carlo simulation and finite-difference methods.

Nonetheless, the tools presented in this report form the mathematical and conceptual foundation of modern quantitative finance. Mastering these concepts is essential for understanding both classical models and their more realistic extensions, and for contributing meaningfully to research and practice in financial engineering.

Appendix

Proof of Itô Isometry

Let H_t be an adapted, square-integrable process. Then the Itô integral

$$\int_0^T H_t dB_t$$

satisfies the identity:

$$\mathbb{E} \left[\left(\int_0^T H_t dB_t \right)^2 \right] = \mathbb{E} \left[\int_0^T H_t^2 dt \right].$$

Sketch of proof: Define the Itô integral first for simple processes (piecewise constant on intervals). Then use isometry for these and extend to all square-integrable H_t by taking limits in L^2 .

Derivation of Black–Scholes PDE

Let $C(t, S_t)$ be the value of a European call option. Apply Itô's formula to $C(t, S_t)$, where:

$$dS_t = rS_t dt + \sigma S_t dW_t.$$

Itô's formula gives:

$$dC = \left(\frac{\partial C}{\partial t} + rS_t \frac{\partial C}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 C}{\partial S^2} \right) dt + \sigma S_t \frac{\partial C}{\partial S} dW_t.$$

Since the price of a derivative must evolve as a martingale under the risk-neutral measure (no-arbitrage), the drift term must match rC . Setting the drift term equal to rC yields the Black–Scholes PDE:

$$\frac{\partial C}{\partial t} + rS \frac{\partial C}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} = rC.$$

Derivation of Geometric Brownian Motion Solution

Consider the SDE:

$$dS_t = \mu S_t dt + \sigma S_t dB_t, \quad S_0 > 0.$$

Define $X_t = \ln S_t$. Then, by Itô's formula:

$$dX_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dB_t.$$

Integrating both sides:

$$X_t = \ln S_0 + \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t,$$

so:

$$S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t \right).$$

Normal Distribution Identities

For standard normal $Z \sim \mathcal{N}(0, 1)$, we have:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad \Phi(x) = \int_{-\infty}^x \phi(u) du.$$

Useful derivative:

$$\frac{d}{dx} \Phi(x) = \phi(x).$$

Greeks Formulas Summary

$$\begin{aligned} \Delta_{\text{call}} &= \Phi(d_1), & \Delta_{\text{put}} &= \Phi(d_1) - 1, \\ \Gamma &= \frac{\phi(d_1)}{S\sigma\sqrt{T-t}}, \\ \text{Vega} &= S\phi(d_1)\sqrt{T-t}, \\ \Theta_{\text{call}} &= -\frac{S\phi(d_1)\sigma}{2\sqrt{T-t}} - rKe^{-r(T-t)}\Phi(d_2), \\ \rho_{\text{call}} &= K(T-t)e^{-r(T-t)}\Phi(d_2). \end{aligned}$$

Appendix: Python Code for Figures

Simulating Brownian Motion Paths

This code generates multiple Brownian motion paths over a time interval.

```
import numpy as np
import matplotlib.pyplot as plt

T = 1
n = 1000
dt = T / n
t = np.linspace(0, T, n)
m = 10 # number of paths

W = np.cumsum(np.sqrt(dt) * np.random.randn(m, n), axis=1)
W = np.insert(W, 0, 0, axis=1)

for i in range(m):
    plt.plot(np.insert(t, 0, 0), W[i])

plt.title("Simulated Brownian Motions")
plt.xlabel("Time")
plt.ylabel("W(t)")
plt.grid(True)
plt.show()
```

Geometric Brownian Motion (GBM) Simulation

This simulates the evolution of stock prices under the GBM assumption.

```
S0 = 100
mu = 0.05
sigma = 0.2

W = np.cumsum(np.sqrt(dt) * np.random.randn(m, n), axis=1)
W = np.insert(W, 0, 0, axis=1)
t = np.insert(t, 0, 0)

S = S0 * np.exp((mu - 0.5 * sigma**2) * t + sigma * W)

for i in range(m):
    plt.plot(t, S[i])

plt.title("GBM Paths for a Stock")
plt.xlabel("Time")
plt.ylabel("S(t)")
```

```
plt.grid(True)
plt.show()
```

Black-Scholes Call Price Calculation

This code computes the theoretical call option price using the Black-Scholes formula.

```
from scipy.stats import norm
import numpy as np

def black_scholes_call(S, K, T, r, sigma):
    d1 = (np.log(S / K) + (r + 0.5 * sigma ** 2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    call_price = S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
    return call_price

S = 100
K = 100
T = 1
r = 0.05
sigma = 0.2

price = black_scholes_call(S, K, T, r, sigma)
print("Black-Scholes Call Price:", round(price, 2))
```

References

References

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- [3] Raluca M. Balan, *Lecture Notes in Stochastic Calculus*, Unpublished lecture notes, University of Ottawa, 2025.